EXCEL

1. Introduction to Excel

Microsoft Excel is a spreadsheet application used for data organization, calculation, analysis, and visualization.
It provides cells arranged in rows and columns that can contain numbers, text, or formulas.

Excel is widely used for:

Data entry and cleaning

Basic statistics and reporting

Financial modeling and business forecasting

Dashboard creation

2. Excel Data Structure

Concept	Description	Example
Workbook	Entire Excel file (.xlsx) containing one or more sheets	SalesReport.xlsx
Worksheet	A single tab (sheet) within a workbook	Sheet1: Sales Data
Cell	Intersection of a row and column	A1, B3, etc.
Range	Group of cells	A1:A10, B2:D5

3. Excel Data Loading and Entry

(a) Manual Data Entry

Directly typing into cells.

(b) Importing Data

Excel supports data import from multiple sources:

CSV / TXT files

Databases (SQL Server, MySQL, etc.)

Web queries and APIs

Power Query (for transformation and automation)

(c) Data Cleaning Tools

Remove Duplicates

Text to Columns

Find & Replace

Data Validation (to control user input)

4. Excel Functions and Formulas

(a) Basic Arithmetic
= A1 + B1
= A2 * 10
= SUM(A1:A10)

(b) Statistical Functions

Function	Description	Example
AVERAGE()	Mean of range	=AVERAGE(B2:B10)
MAX(), MIN()	Highest or lowest value	=MAX(C1:C20)
COUNT()	Counts numeric entries	=COUNT(A1:A10)

(c) Text Functions

Function	Description	Example
CONCATENATE() / &	Combine text	=A1 & " " & B1
LEFT(), RIGHT(), MID()	Extract substrings	=LEFT(A1,3)
UPPER(), LOWER()	Change case	=UPPER(B2)

(d) Logical Functions

Function	Description	Example
IF()	Conditional statement	=IF(A1>50,"Pass","Fail")
AND(), OR()	Multiple conditions	=AND(A1>50,B1<100)

(e) Lookup and Reference Functions

Function	Purpose	Example
VLOOKUP()	Search vertically	=VLOOKUP(A2,Table,3,FALSE)
HLOOKUP()	Search horizontally	=HLOOKUP(B2,Table,2,FALSE)
INDEX() & MATCH()	Flexible lookup	=INDEX(B2:B10,MATCH(E1,A2:A10,0))

5. Excel Charts and Visualization

Excel supports many visualization types:

Column / Bar Charts

Pie Charts

Line Charts

Pivot Charts

Combo Charts

You can customize chart titles, legends, axis labels, and data labels for clarity.

6. Pivot Tables and Dashboards
(a) Pivot Table

A Pivot Table is a tool for summarizing, analyzing, and aggregating large datasets interactively.

Steps:

Select data range

Insert → PivotTable

Drag fields to Rows, Columns, Values, Filters

(b) Pivot Chart

Visual representation of a Pivot Table.

(c) Dashboards

Combine multiple charts, slicers, and pivot tables into an interactive summary view.

EXCEL FORMULAS AND SHORTCUTS

Excel Formula Shortcut Key

1	Shortcut Key	Function	21	Shortcut Key	Function
2	=	Start a formula	22	Ctrl + Shift + @	Apply time format
3	Alt + =	AutoSum selected cells	23	Ctrl + Shift + "%"	Apply percentage format
4	Ctrl + Shift + Enter	Enter an array formula	24	Ctrl + Shift + "^"	Apply exponential format
5	F2	Edit formula in active cell	25	Ctrl + Shift + !	Apply number format with 2 decimals
6	Ctrl + D	Copy formula down	26	Ctrl + Shift + #	Apply date format
7	Ctrl + R	Copy formula right	27	Alt + Enter	Start a new line in the same cell
8	Shift + F9	Calculate selected part of formula	28	Ctrl + Shift + L	Toggle AutoFilter
9	Ctrl + Shift + $	Apply currency format	29	Alt + H, O, I	AutoFit Column Width
10	Ctrl + `	Show all formulas in sheet	30	Ctrl + Shift + &	Apply outline border
11	Ctrl + Shift + Enter	Array formula	31	Ctrl + -	Delete selected row/column
12	Ctrl + ;	Insert today's date	32	Ctrl + Space	Select column
13	Ctrl + Shift + :	Insert current time	33	Shift + Space	Select row
14	F4	Repeat last action / lock cell reference	34	Ctrl + 1	Format Cells dialog
15	Ctrl + K	Insert hyperlink	35	F5	Go To box
16	Ctrl + Shift + U	Expand formula bar	36	Ctrl + G	Go To box
17	Shift + F3	Insert function	37	F8	Extend selection
18	Ctrl + Space	Select column	38	Shift + Space	Select row
19	Ctrl + Shift + L	Toggle AutoFilter	39	Ctrl + Alt + F9	Calculate all worksheets
20	Alt + M, V	Evaluate Formula tool	40	Ctrl + T	Create table from range

TOP 22 Excel formulas

Must Know!

By Nicolas Boucher

Core Calculation Formulas

1. SUM

Adds all numbers in a range

=SUM(A1:A9)

Add values from cell A1 to A9

2. AVERAGE

Calculates the average of numbers in a range

=AVERAGE(B1:B9)

Average of B1 to B9

3. ROUND

Rounds a number to a specified number of digits

=ROUND(C1,2)

Round C1 to 2 decimal places

4. MIN

Identifies the lowest number in a range

=MIN(D1:D9)

Find the minimum in D1 to D9

5. MAX

Finds the highest number in a range

=MAX(D1:D9)

Get the maximum value in D1 to D9

6. COUNT

Counts the number of cells that contain numbers

=COUNT(F1:F9)

Count numeric cells in F1 to F9

*It only counts the numerical inputs

Countblank: Count all the empty cells

Referencing

7. INDEX

Returns a value in table based on row & column number

=INDEX(K1:L10,3,2)

For the value in the 3rd row & 2nd column from the range K1:L10

8. MATCH

Searches for a value and returns its relative position

=MATCH(L1,M1:M9,0)

to find L1's position from the range M1:M9

9. VLOOKUP

Searches for a value in the first column of a range

=VLOOKUP (G1,H1:I10,2,FALSE)

To find G1 in H1:H10

2 0columns containing the data need to be retrieved

Tips

- For an exact match => FALSE
- Upwards matching => TRUE (useful when the look up value is a number)

10. HLOOKUP

Searches for a value in the top row of a table.

=HLOOKUP (J3,J1:O2,2,FALSE)

To find J3 in the top row J1:O1

11. XLOOKUP

Finds a value in one range & returns a result from another

=XLOOKUP (A7,A2:A4,B2:C4)

To find A7 in A2:A4 & return corresponding value from B2:C4 if a match is found

Conditional Calculations

12. COUNTIF / COUNTIFS

Counts cells that meet a condition

=COUNTIF (N1:N9,">5")

To count cells greater than 5

13. COUNTA

Counts non-empty cells

=COUNTA(O1:O9)

To count all non-empty cells in a range

14. SUMIF / SUMIFS

Adds cells that meet a condition

=SUMIF (P1:P9,">5")

To sum values over 5

15. IF / NESTED IFS

Performs conditional tests

=IF(R1>5,"Yes","No")

For simple conditions

Note

- IF: 1 variable
- IFS: multiple variables

=IF(test, true, false)

=IF(test1, true, IF(test2, true, false)) → Nested IF

=IFS(test1, true, test2, true, test3, true)

16. AVERAGEIF / AVERAGEIFS

Calculates the average for cells that meet criteria

=AVERAGEIF (Q1:Q9,">5")

Average of values over 5

17. Logical Operators (AND, OR)

Combines multiple conditions

=AND(S1>5,S2<2) — Both conditions are true => Output is true

=OR(S1>5,S2<2) — Just 1 condition is true => Output is true

Output Format Management

18. CONCATENATE

Joins two or more text strings

=CONCATENATE(K1," ",L1)

Combine first name and last name in one cell

19. LEFT / RIGHT

Extracts characters from the start/end of a text string

=LEFT(U1,5) — The result:"Excel"

=RIGHT(U1,5) — The result: "Class"

20. MID

Extracts characters from the middle of a text string

=MID(U1,7,7)

7 characters starting from the 7th => Mastery

21. TEXT

Converts a value to text in a specific number format

=TEXT(V1,"MMM-YY")

17/02/2024 => Feb-24

22. IFERROR

Returns a custom result if a formula errors out

=IFERROR(W1,"0")

Give 0 as output, instead of #div error

Follow Nicolas Boucher

POWER BI